

Task 3 Write-Up: Adversity Prediction Model

Model Choice: Gradient Boosted Trees (GBT)

We use **sklearn GradientBoostingClassifier** with one model trained per horizon τ . GBTs were chosen over alternatives for the following reasons:

- **Non-linear feature interactions:** Adversity prediction requires capturing interactions between client identity, spread, inventory pressure and time-of-day — GBTs handle these naturally.
- **Robustness to outlier trades:** Volume and spread distributions are heavy-tailed; tree-based splits are invariant to monotone feature transformations.
- **Calibrated probabilities:** GBT with sufficient depth produces well-calibrated class probabilities, essential for the externalization decision in Task 4.
- **No temporal leakage:** All training uses only features observable at trade time (no future mid-prices). The train/val/test split is strictly chronological (60/20/20 by date).

Hyperparameters: 150 estimators, max_depth=4, learning_rate=0.05, subsample=0.8, min_samples_leaf=20. The moderate depth prevents overfitting to noise in individual short-horizon label realisations.

Feature Vector (14 features)

Index	Feature	Rationale
0	side	Trade direction; buy-side and sell-side adversity differ systematically
1	log_volume	Large trades carry more information (Kyle λ); log compresses tail
2	spread	Wider spread $\rightarrow$ more adverse-selection compensation priced in
3	TP – M0	Proxy for quoted half-spread; captures LP pricing aggressiveness
4	spread / M0	Dimensionless relative spread; comparable across price levels
5	η (eta)	Elapsed day fraction; intraday patterns in informed trading
6	momentum	5-trade M0 return; recent trend may predict continuation
7	realized_vol σ	RMS of last-20 mid returns (eq. 10); risk regime signal
8–13	client_A ... client_F	Client one-hot; each client has distinct adversity behaviour

Table 3: Feature vector definition and rationale.

Data Split

Date range: 3 Mar 2025 – 29 Apr 2025 (~40 trading days). Split: train days 1–24 (60%), validation days 25–32 (20%), test days 33–40 (20%). Chronological ordering prevents any look-ahead bias. Features are standardised (zero mean, unit variance) per split using the training scaler.

Performance Metrics

Split	Accuracy	Precision	Recall	Log Loss
Train	0.567	0.596	0.254	0.679
Validation	0.557	0.519	0.243	0.685
Test	0.558	0.526	0.242	0.685

Table 4: Averaged metrics across all six horizons ($\tau=5, \dots, 30$).

Interpretation and Limitations

Accuracy of ~56% is modest but consistent with the literature: short-horizon adversity prediction is fundamentally hard because price moves at 5–30 second horizons carry high noise. The model generalises well — val and test metrics are within 1% of training, indicating no overfitting.

The low recall (~24%) reflects the model's conservative bias: it flags adverse trades only when confidence is high. For externalization (Task 4), this means the strategy externalizes fewer trades but with higher precision. Precision of ~53% on test set means that when the model does flag a trade as adverse, it is correct more than half the time.

Opportunities for improvement: Per-client models would capture individual adversity patterns more precisely (F's profile is very different from A's). Additional features such as rolling order-flow imbalance, inter-arrival time between trades, and spread-to-vol ratio could materially improve recall.